1. What percentage of 40 is 8?

   (A) 5
   (B) 20
   (C) 32
   (D) 150

2. The exact value of $6 \div (0.0003)$ is

   (A) 200
   (B) 2 000
   (C) 20 000
   (D) 200 000

3. If $6 \times 10^{-n} = 0.06$, $n > 0$, then n is

   (A) 1
   (B) 2
   (C) 3
   (D) 4

4. Paula bought $1\frac{7}{8}$ m of fabric to make a shirt, while Rita bought $1\frac{3}{4}$ m to make her shirt. How much MORE fabric did Paula buy than Rita?

   (A) $\frac{1}{8}$ m
   (B) $\frac{1}{2}$ m
   (C) $1\frac{1}{8}$ m
   (D) $3\frac{5}{8}$ m

5. The value of 29.94×0.5 is approximately

   (A) 0.15
   (B) 1.5
   (C) 15
   (D) 150

6. There are 40 students in a class. Girls make up 60% of the class. 25% of the girls wear glasses. How many girls in the class wear glasses?

   (A) 6
   (B) 8
   (C) 10
   (D) 15

7. $(3 \times 7) + (3 \times 5)$ is the same as

   (A) $3 \times 5 + 7$
   (B) $3 \times 7 + 5$
   (C) $3 \times (7 + 5)$
   (D) $(3 + 7) \times 5$

8. What is the HIGHEST common factor of the set of numbers $\{30, 45, 90\}$?

   (A) 3
   (B) 5
   (C) 15
   (D) 90

9. What is the LEAST number of plums that can be shared equally among either 6, 9 or 12 children?

   (A) 27
   (B) 36
   (C) 54
   (D) 72

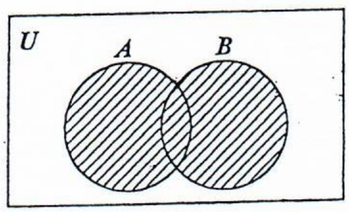
10. If $3n$ is an odd number, which of the following is an even number?

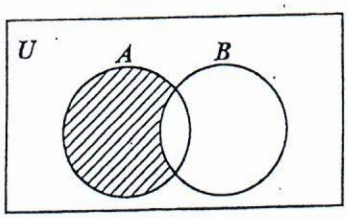
    (A) $3n - 1$
    (B) $3n + 2$
    (C) $3n - 2$
    (D) $3n + 2n$

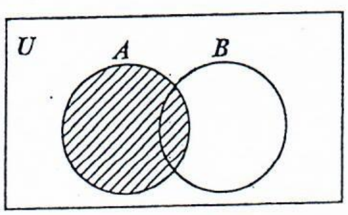
11. If $A = \{3, 6, 9\}$, then the number of subsets of A is

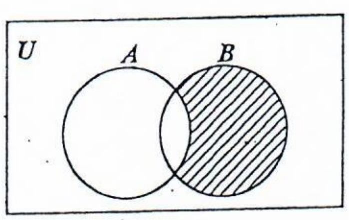
(A) 2
(B) 3
(C) 4
(D) 8

12. In which of the following Venn diagrams is the region $A \cap B'$ shaded?

(A)


(B)


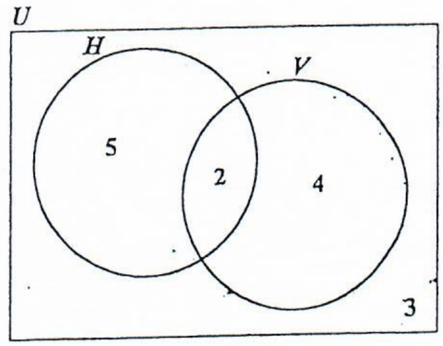
(C)


(D)


13. If the universal set $U = \{1, 2, 3, 4, 5, 6\}$ and $H = \{3, 4, 6\}$, then $H' =$

(A) $\{3, 4, 6\}$
(B) $\{1, 3, 5\}$
(C) $\{2, 4, 6\}$
(D) $\{1, 2, 5\}$

Item 14 refers to the following Venn diagram.



14. In the Venn diagram

$U =$ {students who play games}
$H =$ {students who play hockey}
$V =$ {students who play volleyball}

The number of students in each set is indicated. How many students do NOT play volleyball?

(A) 2
(B) 3
(C) 5
(D) 8

15. The simple interest on \$600 for 4 years at 5 per cent per annum is

(A) $\dfrac{\$600 \times 5 \times 4}{100}$

(B) $\dfrac{\$100 \times 5 \times 4}{600}$

(C) $\dfrac{\$600 \times 5}{100 \times 4}$

(D) $\dfrac{\$100 \times 4}{600 \times 5}$

16. $3\frac{1}{4}\%$ of \$500 is

(A) \$ 1.62
(B) \$15.52
(C) \$16.00
(D) \$16.25

17. Mrs Jones has US\$200 to change to TT\$. The bank gives her TT\$6.20 for US\$1 and charges TT\$2.50 for the transaction. Mrs Jones receives

(A) TT\$1 237.50
(B) TT\$1 238.50
(C) TT\$1 240.00
(D) TT\$1 242.50

18. A man's basic wage for a 40-hour week is \$160.00. He is paid \$5.00 per hour for overtime. If he works $6\frac{1}{2}$ hours overtime in a certain week, his wage for that week is

(A) \$165.00
(B) \$166.50
(C) \$171.50
(D) \$192.50

19. A dinner at a restaurant was advertised at \$60 plus 18% tax. The TOTAL bill for this dinner was

(A) \$60.00
(B) \$70.80
(C) \$78.00
(D) \$81.60

20. A store offers a discount of 10% to customers who spend more than \$20. If a customer's total bill is \$80, what will he actually pay?

(A) \$60
(B) \$70
(C) \$72
(D) \$74

21. A salesman sells a car for \$10 500. If he is paid a commission of 5% for the first \$10 000 and 8% on the remainder, then the TOTAL commission he receives is

(A) \$525
(B) \$540
(C) \$545
(D) \$565

22. Mr Jones buys a car for \$64 000. The car depreciates by 20% in the first year and 10% in each of the following years. The value of the car at the end of the second year is

(A) \$19 200
(B) \$44 800
(C) \$46 080
(D) \$51 200

23. $15p + 55q - 2(3p + 4q) =$

(A) $9p + 47q$
(B) $9p + 63q$
(C) $21p + 63q$
(D) $70p - 14q$

24. $-(-2q) - 3q =$

(A) $-6q$
(B) $-5q$
(C) $-q$
(D) $5q$

25. $2^6 \div 2^2 =$

(A) 2^3
(B) 2^4
(C) 2^8
(D) 2^{12}

26. If $\frac{p}{5} = 20$, then $p =$

(A) $20 - 5$
(B) $20 + 5$
(C) $20 \div 5$
(D) 20×5

27. "When 7 is added to 3 times a certain number, n, the result is 22."

The statement above may be represented by the equation

(A) $7n - 22 = 3$
(B) $3n + 22 = 7$
(C) $3n + 7 = 22$
(D) $7n + 3 = 22$

28. Thirty-six men can paint a shopping mall in 25 days. How many days will it take 15 men to paint the mall, if they work at the same rate?

(A) 50
(B) 56
(C) 60
(D) 72

29. If $x = 4$ and $y = 2$, then the value of

$\frac{x^2 + 3y}{xy}$ is

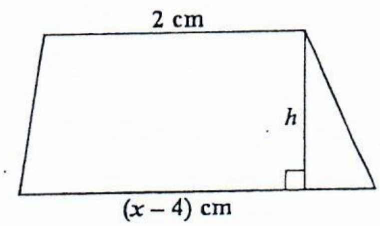
(A) $1\frac{3}{4}$

(B) $2\frac{1}{2}$

(C) $2\frac{3}{8}$

(D) $2\frac{3}{4}$

Item 30 refers to the following diagram of a trapezium.



30. The trapezium with height, h, has an area of x^2 cm^2. The equation that may be used to find the value of x is

(A) $x^2 = \frac{h}{2}(x - 2)$

(B) $x^2 = h(x - 1)$

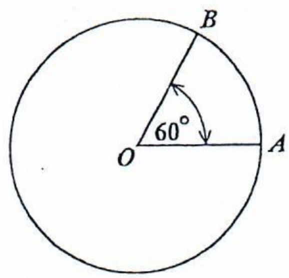
(C) $x^2 = \frac{h}{2}(x - 6)$

(D) $x^2 = h(x - 4)(x + 2)$

31. Given that $3(x-1)-2(x-1)=7$, the value of x is

(A) 6
(B) 7
(C) 8
(D) 9

Item 32 refers to the following diagram of a circle.



32. In the circle above, with centre O, the area is 20 cm^2. The area of the minor sector AOB, in cm^2, is

(A) $\frac{60}{360} \times 20$

(B) $\frac{120}{360} \times 20$

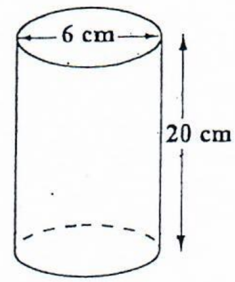
(C) $\frac{60}{360} \times 20 \times 20$

(D) $\frac{120}{360} \times 20 \times 20$

33. 2 500 millimetres expressed in metres is

(A) 0.25
(B) 2.5
(C) 25
(D) 250

Item 34 refers to the following diagram which shows a cylinder with diameter 6 cm and height 20 cm.



34. The volume, in cm^3, of the cylinder is

(A) 180π
(B) 240π
(C) 360π
(D) 720π

35. The distance around the edge of a circular pond is 88 m. The radius, in metres, is

(A) 88π

(B) 176π

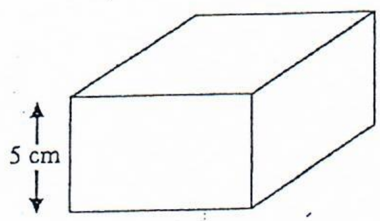
(C) $\frac{88}{\pi}$

(D) $\frac{44}{\pi}$

36. An aircraft leaves A at 16:00 hours and arrives at B at 19:30 hours, travelling at an average speed of 550 kilometres per hour. A and B are in the same time zone. The distance from A to B, in kilometres, is

(A) 907.5
(B) 962.5
(C) 1815
(D) 1925

Item 37 refers to the following diagram, **not drawn to scale**, which shows a cuboid.



37. The volume of the cuboid is 320 cm^3 and the height is 5 cm. If the cuboid has a square base, what is the length of one side of the base?

(A) 8 cm
(B) 16 cm
(C) 32 cm
(D) 64 cm

38. A square has the same area as a rectangle with sides of length 9 cm and 16 cm. What is the length of the side of the square?

(A) 9 cm
(B) 12 cm
(C) 12.5 cm
(D) 75 cm

39. The distance around a lake is 8 km. On a map, this distance around the lake is represented by a length of 2 cm. The scale on the map is

(A) 1 : 40
(B) 1 : 2 000
(C) 1 : 200 000
(D) 1 : 400 000

40. The median of the numbers

1, 1, 5, 5, 6, 7, 7, 7, 7, 8 is

(A) 7
(B) 6.5
(C) 6
(D) 5.4

Item 41 refers to the following table which shows the frequency of scores obtained by students in a test.

Scores	2	3	5	6	8	11
Students	8	4	6	3	12	2

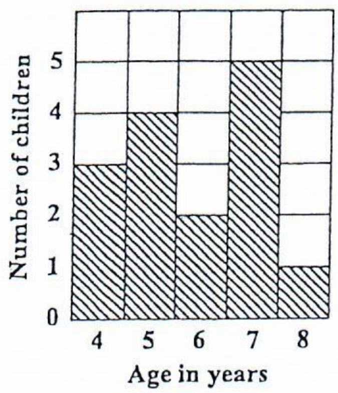
41. The modal score is

(A) 8
(B) 9
(C) 10
(D) 12

42. The heights, in cm, of 10 students are 150, 152, 155, 153, 170, 160, 156, 165, 158, 155. The range is

(A) 5
(B) 20
(C) 150
(D) 155

Item 43 refers to the following bar chart which shows the ages of children who took part in a survey.



43. How many children took part in the survey?

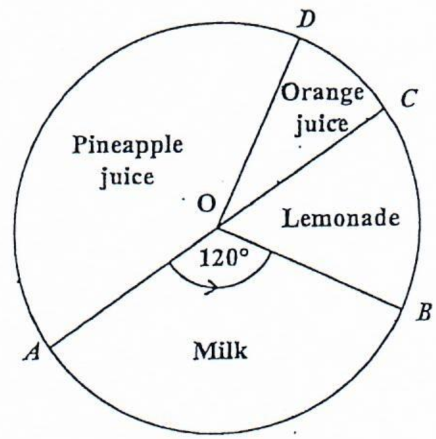
(A) 5
(B) 15
(C) 75
(D) 87

44. Six hundred students write an examination. The probability of a randomly selected student failing the examination is $\frac{1}{5}$.

How many students are expected to **pass**?

(A) 100
(B) 120
(C) 480
(D) 500

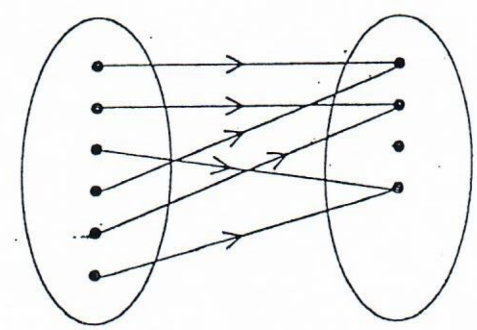
Item 45 refers to the following piechart which shows the drinks preferred by a group of students. AOC is a diameter of the circle.



45. If 12 students prefer lemonade, then the TOTAL number of students is

(A) 48
(B) 72
(C) 180
(D) 360

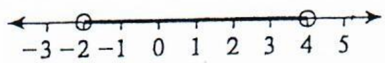
Item 46 refers to the following mapping diagram.

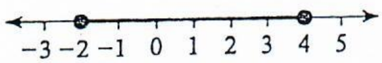


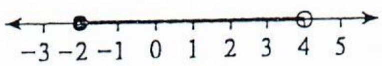
46. The relationship that BEST describes the mapping in the diagram is

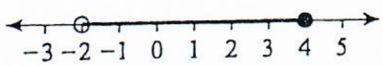
(A) one-to-one
(B) one-to-many
(C) many-to-one
(D) many-to-many

47. Which of the following line graphs represents $\{ x : -2 < x \leq 4 \}$?

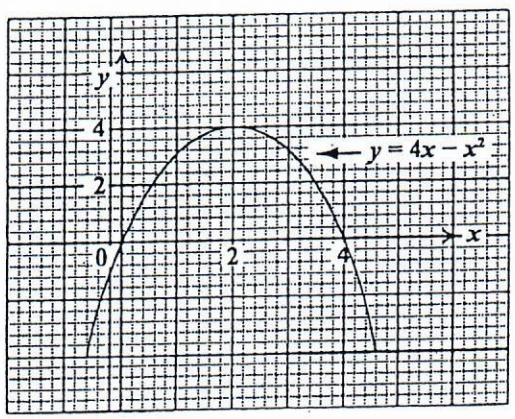
(A) 

(B) 

(C) 

(D) 

Items 48–49 refer to the following graph of a quadratic function.



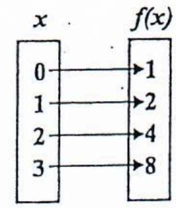
48. The maximum point of $y = 4x - x^2$ is

(A) (0, 0)
(B) (0, 4)
(C) (2, 4)
(D) (4, 2)

49. The values of x at the points where $y = 4x - x^2$ intersects $y = 0$ are

(A) $x = 0$ and $x = 4$
(B) $x = 0$ and $x = 2$
(C) $x = 2$ and $x = 4$
(D) $x = 0$ and $x = -4$

Item 50 refers to the following mapping diagram.



50. The diagram represents the mapping

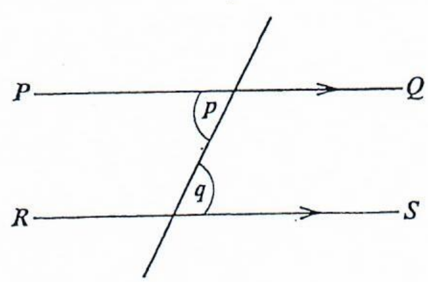
(A) $f(x): x \rightarrow x + 2$
(B) $f(x): x \rightarrow x + 1$
(C) $f(x): x \rightarrow 2^x$
(D) $f(x): x \rightarrow 2x$

51. Which of the following sets is represented by the function $f: x \rightarrow x^2 + 3$ where

$x \in \{0, 1, 2, 3\}$?

(A) $\{(0, 3), (1, 1), (2, 4), (3, 9)\}$
(B) $\{(0, 3), (1, 4), (2, 5), (3, 6)\}$
(C) $\{(0, 3), (1, 5), (2, 7), (3, 9)\}$
(D) $\{(0, 3), (1, 4), (2, 7), (3, 12)\}$

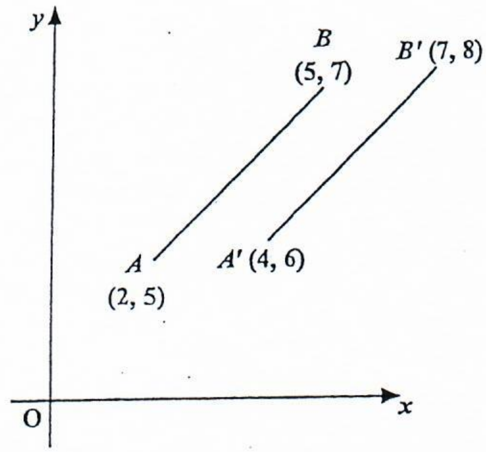
Item 52 refers to the following transversal diagram.



52. In the diagram PQ and RS are parallel. Which of the following BEST describes the relation between p and q?

(A) $p = q$
(B) $p < q$
(C) $p + q = 90°$
(D) $p + q = 180°$

Item 53 refers to the following diagram which shows a translation.



53. In the diagram, the translation by which AB is mapped onto $A'B'$ is represented by

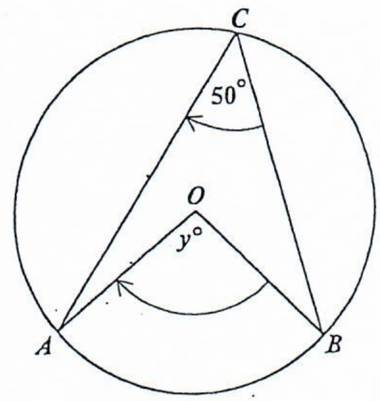
(A) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

(C) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

(D) $\begin{pmatrix} 5 \\ 3 \end{pmatrix}$

Item 54 refers to the following diagram of a circle.

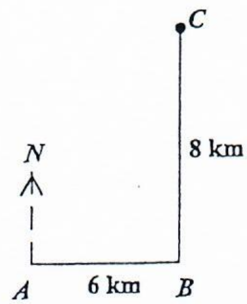


54. If O is the centre of the circle, then $y°$ is

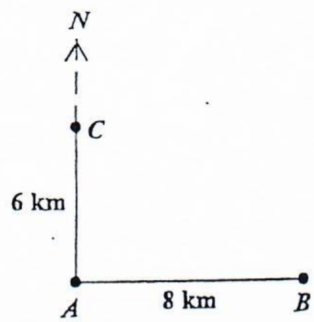
(A) 25°
(B) 80°
(C) 90°
(D) 100°

55. A ship sailed 8 km due east from A to B. It then sailed 6 km due north to C. Which of the following diagrams BEST represents the path of the ship?

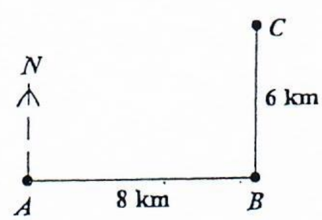
(A)



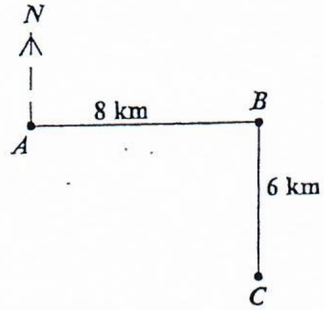
(B)



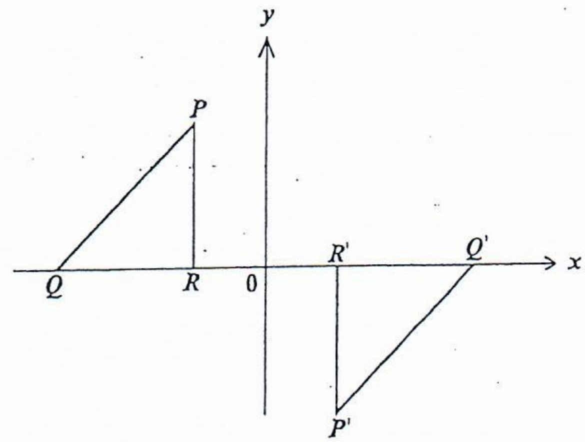
(C)



(D)



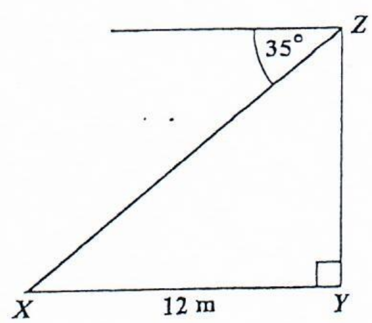
Item 56 refers to the following diagram which shows a transformation.



56. The transformation which will map triangle PQR onto $P'Q'R'$ is

(A) a rotation of 180° about the origin
(B) a reflection in the line $y = x$
(C) an enlargement, centre R, of scale factor -1
(D) a translation parallel to the x-axis

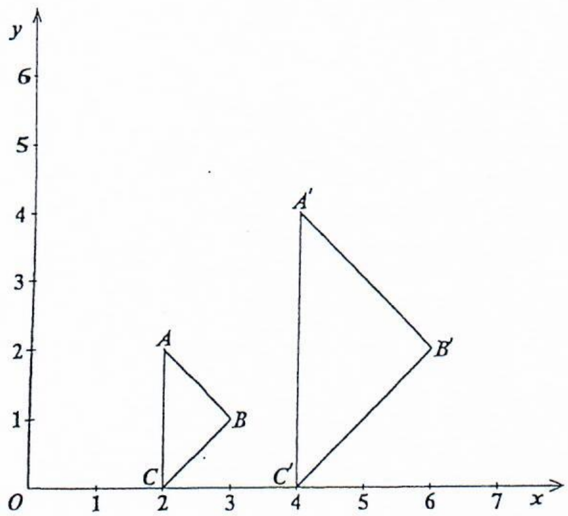
Item 57 refers to the following diagram.



57. The diagram, **not drawn to scale**, shows that the angle of depression of point X from Z is 35°. If X is 12 metres from Y, the height YZ, in metres, is

(A) 12 cos 35°
(B) 12 sin 35°
(C) 12 tan 35°
(D) 12 cos 55°

Item 58 refers to the following diagram which shows an enlargement.



58. In the diagram, triangle ABC is mapped onto triangle $A'B'C'$ where O is the centre of enlargement. What is the scale factor of the enlargement?

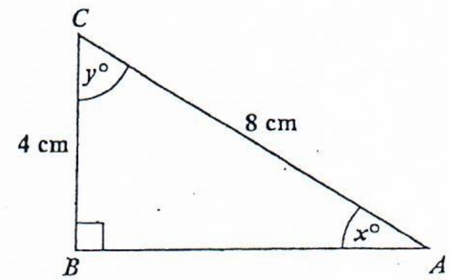
(A) $\frac{1}{2}$

(B) -2

(C) $-\frac{1}{2}$

(D) 2

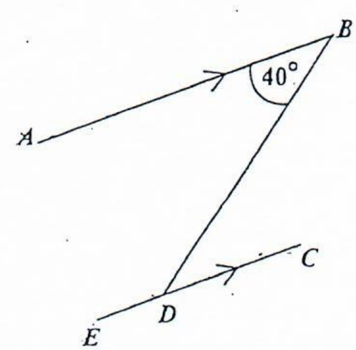
Item 59 refers to the following diagram of a triangle.



59. In the right-angled triangle above, which trigonometric ratio is equal to $\frac{4}{8}$?

(A) $\sin x$
(B) $\tan y$
(C) $\cos x$
(D) $\tan x$

Item 60 refers to the following diagram.



60. In the diagram above, AB is parallel to EC. The value of $\angle BDE$ is

(A) 40°
(B) 50°
(C) 120°
(D) 140°

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

SUBJECT: ______ PROFICIENCY: ______ 2018. CANDIDATE NAME: ______

CARIBBEAN EXAMINATIONS COUNCIL

MULTIPLE CHOICE
ANSWER SHEET

A REGISTRATION NUMBER

C TEST CODE

D ABSENT

E

GRIDDING OCCURRENCES	1	2	3	4
GRIDDED INITIALS				
CHECKED INITIALS				
SCANNING OCCURRENCES	1	2	3	4

SIGNATURE OF CANDIDATE

AT THE END OF THE EXAMINATION PLACE THIS ANSWER SHEET BETWEEN THE MIDDLE PAGES OF THE QUESTION PAPER BOOKLET THEN SEAL THE TOP, BOTTOM AND FORE EDGE OF THE BOOKLET USING THE THREE SEALS PROVIDED.

B MARK YOUR ANSWERS BELOW

1 A B C D	11 A B C D	21 A B C D	31 A B C D	41 A B C D	51 A B C D
2 A B C D	12 A B C D	22 A B C D	32 A B C D	42 A B C D	52 A B C D
3 A B C D	13 A B C D	23 A B C D	33 A B C D	43 A B C D	53 A B C D
4 A B C D	14 A B C D	24 A B C D	34 A B C D	44 A B C D	54 A B C D
5 A B C D	15 A B C D	25 A B C D	35 A B C D	45 A B C D	55 A B C D
6 A B C D	16 A B C D	26 A B C D	36 A B C D	46 A B C D	56 A B C D
7 A B C D	17 A B C D	27 A B C D	37 A B C D	47 A B C D	57 A B C D
8 A B C D	18 A B C D	28 A B C D	38 A B C D	48 A B C D	58 A B C D
9 A B C D	19 A B C D	29 A B C D	39 A B C D	49 A B C D	59 A B C D
10 A B C D	20 A B C D	30 A B C D	40 A B C D	50 A B C D	60 A B C D

IF THERE ARE MORE THAN 60 QUESTIONS TURN OVER AND CONTINUE ON THE OTHER SIDE